[Interview] Solutions Engineer - Take Home Assignment

Objective

Part of your job is building prototypes to show to our customers. Oftentimes, we need to turn these around in a day or two. Your task is to build a customer-ready prototype to show to an insurance company.

Build a minimal full-stack prototype that:

1. Allows a user to **upload** (or provide a URL of) a photograph of a damaged car.
2. Generates the following outputs:
    - **Car Metadata**: Make, model, and color.
    - **Damage Summary**: A textual description of the damage (e.g., "Left side rear bumper dent with scratch").
    - **Estimated Repair Cost**: Provide a rough, AI-generated cost estimate.

Tools You Can Use

Anything! Use any libraries/services/AI tools you want to complete this task at a high quality and within the timeframe given.

High-Level Requirements

1. **Front-End**
    - A simple UI (e.g., a single-page web interface or a minimal React/Next page) that allows the user to:
        - Upload a car photo (or paste an image URL).
        - See the AI-generated results (metadata, damage summary, cost estimate) displayed on the same page.
2. **Back-End/API**
    - Use whatever backend you'd like. It just needs to be able to process an image and provide the metadata, summary, and cost.
3. **Time Constraint**
    - Aim to complete this in **3–4 hours**. We want to see how you prioritize tasks and prototype efficiently. If you can get more done that you think would impress the customer, that's great but don't spend too much time on this. The idea is to see what you could turn around in a day.

What to Deliver

We will walk through your solution live in an interview. Come prepared to with the following:

1. **Working Demo**
    - A link to a deployed version (if possible) or instructions on how to run it locally.

    - Show that the user can upload a photo or paste a URL and get back results on the same page.
2. **Code Repository + README**
    - Host on GitHub/GitLab/Bitbucket.
    - **README** should include:
        - **Setup Instructions** (How to install dependencies, start the app).
        - **Architecture Overview** (Which tools/APIs you used, high-level flow of data).
3. **Brief Design Explanation**
    - Either in your README or a short video (like Loom), explain:
        - **Why you chose certain tools** (e.g., Bolt.new for rapid front-end/back-end building).
        - **How your AI logic works**
        - **Potential improvements** if you had more time (e.g. database to store history, user authentication, etc.).
4. **Statement of Work (SOW) Development**
    - Create a comprehensive SOW document (3 pages maximum) that includes:
        - **Project Scope:** Define the boundaries of the AI-powered claims management solution
        - **Technical Approach:** Detail the technical implementation strategy, including AI/ML models and integration points
        - **Milestones and Timeline:** Outline key deliverables and realistic timeframes
        - **Risk Assessment:** Identify potential technical challenges and mitigation strategies
        - **Success Metrics:** Define clear KPIs to measure project success

Key Evaluation Points

- **End-to-End Thinking**: Do you demonstrate how front-end, back-end, and AI all fit together?
- **Rapid Prototyping**: Can you deliver a functional prototype quickly?
- **Clarity**: Is your code and documentation easy to understand?
- **Creativity & Practicality**: How would you scale or enhance your solution?

Feel free to expand the scope if you have extra time—just be sure to meet the main requirements and keep it within the 3–4 hour scope. Good luck, and we look forward to seeing what you build!